

OTP - One Time Pad

➔ Improvement over Vigenere cipher

Algorithm : combine the message and the key

Key : an infinite random string

Key space : 26^n (n being the length of the plaintext)

	whatanicedaytoday
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Advantage : **this is the perfect cipher !**

Disadvantage : hard to use in practice, how to transmit the key ?

The impossibility of breaking OTP

The ciphertext bears no statistical relationship to the plaintext

➡ No statistical analysis

For any plaintext and ciphertext, there exists a key mapping one to the other, and all keys are equally probable

➡ A ciphertext can be decrypted to any plaintext of the same length